

In the context of the on-demand economy, such decisions may also have to be taken rapidly as the time scale from a customer placing an order online to expecting delivery grows ever shorter. The time available for running (and re-running) algorithms grows shorter; this becomes acute if a last-minute change of plan is called for due to issues such as staff shortages, weather or vehicle availability.

6.2 Illumination Algorithms—MAP-Elites

As discussed in Chap. 4 the construction of a non-dominated front gives the user a choice of solutions, this has the advantage of incorporating the users' expertise and allowing for a level of decision-making above the planning undertaken by the algorithm. This principle may be extended through the use of an *illumination algorithm* (Mouret and Clune 2015). An Illumination Algorithm seeks to find a set of high-quality solutions that represent the entire solution space, giving the user a large number of solutions to choose from.

The Map Archive of Phenotypic Elites (MAP-Elites) developed by Mouret et al. (2015) seeks to build a *map of phenotypic elites*, in simple terms, the algorithm seeks to build a structured set of high-quality solutions that the user can choose from representing possible solutions from across the entire search space. The key points are that the solutions are held in a structure which allows a user to “browse” through them and that the solutions are *elite*, that is to say they represent high-quality (or in real-world terms, useful) solutions.

Consider a logistics problem that has 2 solution characteristics, financial cost c and environmental impact (CO_2) e . Suppose we evolved the following solutions to